function down as an attacker has no chance of trying all possible session tokens.

> **DEFINITION** A ***hash function*** takes an arbitrarily-sized input and produces a fixed-size output. A hash function is ***cryptographically secure*** if it is infeasible to work out what input produced a given output without trying all possible inputs (known as *preimage resistance*), or to find two distinct inputs that produce the same output (*collision resistance*).

